

23CSE472 – AI & Robotics Lab

Worksheet – 1

Introduction to Robotics Toolbox & Basic Pose Representation

1. Creating rotation matrix in 2d and 3d

2d Rotation Function:

$R = \text{ROT2}(\text{THETA})$ THETA is in radian

$R = \text{ROT2}(\text{THETA}, 'deg')$ as above but THETA is in degrees

```
>>R = rot2(45,'deg')  
  
R =  
  
    0.7071    -0.7071  
    0.7071     0.7071  
  
>> R1 = rot2(0.7853)  
  
R1 =  
  
    0.7072    -0.7070  
    0.7070     0.7072
```

3d Rotation Function:

$R = \text{ROTX}(\text{THETA}), \text{ROTY}(\text{THETA}) \& \text{ROTZ}(\text{THETA})$ THETA is in radian

$R = \text{ROTX}(\text{THETA}, 'deg')$ as above but THETA is in degrees. Similary for Y & Z axis.

```
>> R = rotx(60,'deg')  
  
R =  
  
    1.0000         0         0  
         0     0.5000    -0.8660  
         0     0.8660     0.5000
```

```
>> R1 = rotx(1.0472)

R1 =

    1.0000    0    0
    0    0.5000   -0.8660
    0    0.8660    0.5000
```

2. Visualizing 2d and 3d frame

Function for 2d

TRPLOT2(T, OPTIONS) draws a 2D coordinate frame represented by the SE(2) homogeneous transform T (3×3).

H = TRPLOT2(T, OPTIONS) as above but returns a handle.

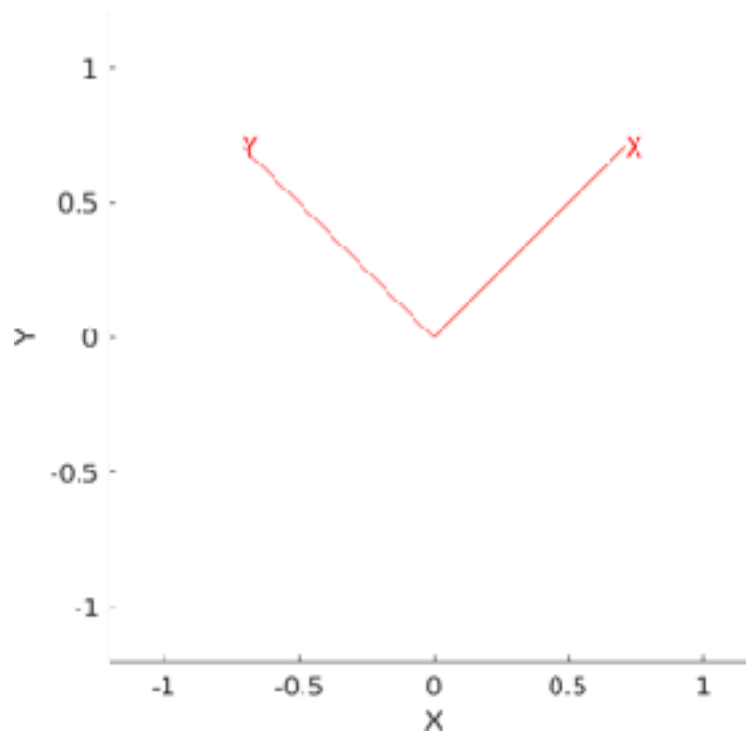
TRPLOT(R, OPTIONS) as above but the coordinate frame is rotated about the origin according to the orthonormal rotation matrix R (2×2).

H = TRPLOT(R, OPTIONS) as above but returns a handle.

H = TRPLOT2() creates a default frame EYE(2,2) at the origin and returns a handle.

```
>>R = rots(45,'deg')

>>trplot2(R, 'color', 'r')
```



Function for 3d

TRPLOT(T, OPTIONS) draws a 3D coordinate frame represented by the SE(3) homogeneous transform T (4×4).

H = TRPLOT(T, OPTIONS) as above but returns a handle.

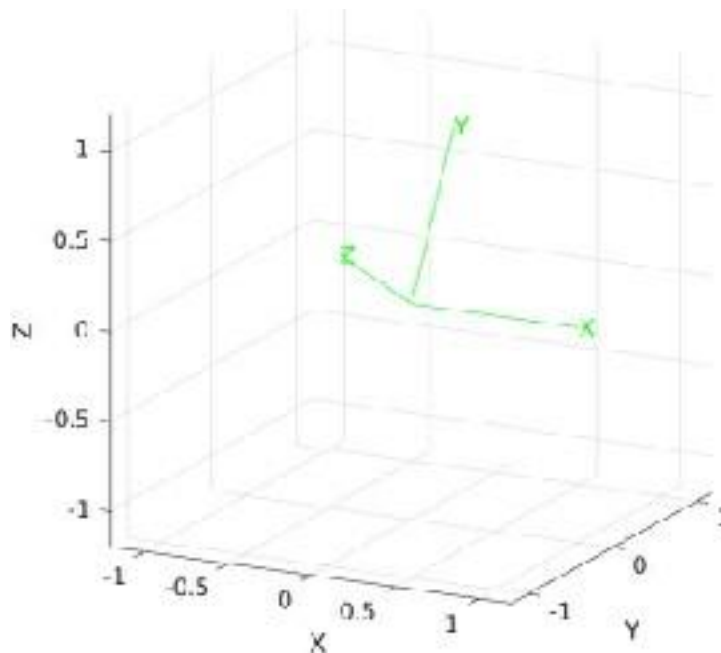
TRPLOT(R, OPTIONS) as above but the coordinate frame is rotated about the origin according to the orthonormal rotation matrix R (3×3).

H = TRPLOT(R, OPTIONS) as above but returns a handle.

H = TRPLOT() creates a default frame EYE(3,3) at the origin and returns a handle.

```
>> R = rotx(1.047)
```

```
>>trplot(R,'color','g')
```



3. Translation in 2d and 3d

Function for 2d

T = TRANSL2(X, Y) is an SE(2) homogeneous transform (3×3) representing a pure translation.

T = TRANSL2(P) is a homogeneous transform representing a translation or point P=[X,Y]. P ($M \times 2$) represents a sequence and T ($3 \times 3 \times M$) is a sequence of homogeneous transforms such that T(:, :, i) corresponds to the i'th row of P.

```
>> T = transl2(2,3)

T =

     1     0     2
     0     1     3
     0     0     1
```

Function for 3d

$T = \text{TRANSL}(X, Y, Z)$ is an $\text{SE}(3)$ homogeneous transform (4×4) representing a pure translation of X , Y and Z .

$T = \text{TRANSL}(P)$ is an $\text{SE}(3)$ homogeneous transform (4×4) representing a translation of $P=[X,Y,Z]$. P ($M \times 3$) represents a sequence and T ($4 \times 4 \times M$) is a sequence of homogeneous transforms such that $T(:, :, i)$ corresponds to the i 'th row of P .

```
>> T1 = transl(1,4,6)

T1 =

     1     0     0     1
     0     1     0     4
     0     0     1     6
     0     0     0     1
```

4. Creating homogenous transformation matrix in 2d and 3d

Function for 2d

$T = \text{TROT2}(\text{THETA})$ is a homogeneous transformation (3×3) representing a rotation of THETA radians.

$T = \text{TROT2}(\text{THETA}, 'deg')$ as above but THETA is in degrees.

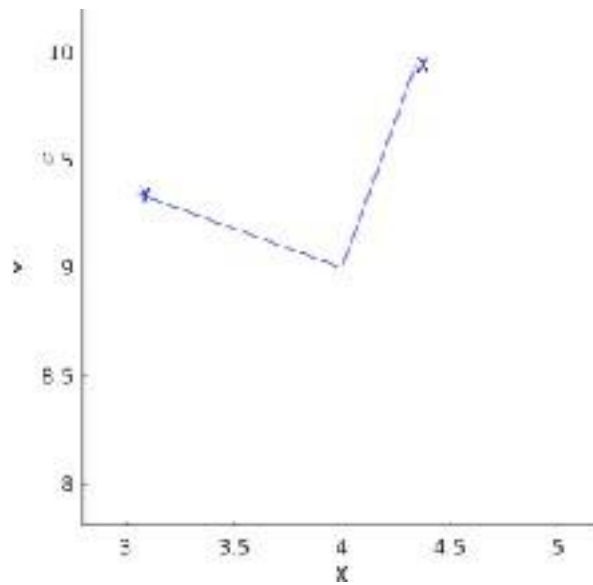
Check out: RT2TR , TR2RT , R2T and T2R

```
>> H = transl2(4,9) * trot2(70,'deg')

H =

    0.3420   -0.9397    4.0000
    0.9397    0.3420    9.0000
         0         0    1.0000
```

```
>> trplot2(H)
```



Function for 3d

$T = \text{TROTX}(\text{THETA}), \text{TROTY}(\text{THETA}), \text{TROTZ}(\text{THETA})$ is a homogeneous transformation (4×4) representing a rotation of THETA radians about the x-axis.

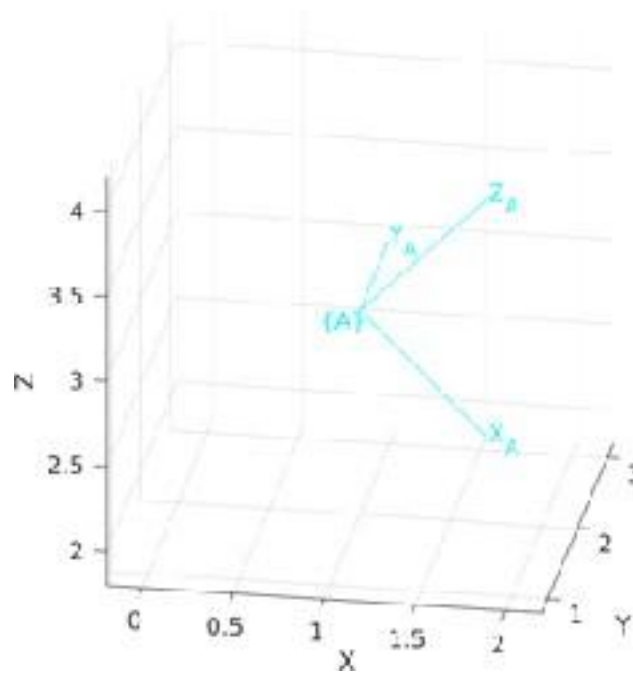
$T = \text{TROTX}(\text{THETA}, 'deg')$ as above but THETA is in degrees. Similarly for Y and Z axis

```
>> H1 = transl(1,2,3) * troty(45,'deg')
```

```
H1 =
```

```
    0.7071         0    0.7071    1.0000
         0    1.0000         0    2.0000
   -0.7071         0    0.7071    3.0000
         0         0         0    1.0000
```

```
>> trplot(H1, 'frame','A','color','c')
```



5. Frame Animation

Function for 2d

TRANIMATE2(P1, P2, OPTIONS) animates a 3D coordinate frame moving from pose X1 to pose X2. Poses X1 and X2 can be represented by:

- SE(2) homogeneous transformation matrices (3×3)
- SO(2) orthonormal rotation matrices (2×2)

TRANIMATE2(X, OPTIONS) animates a coordinate frame moving from the identity pose to the pose X represented by any of the types listed above.

TRANIMATE2(XSEQ, OPTIONS) animates a trajectory, where XSEQ is any of

- SE(2) homogeneous transformation matrix sequence ($3 \times 3 \times N$)
- SO(2) orthonormal rotation matrix sequence ($2 \times 2 \times N$)

```
>> H = transl2(4,9) * trot2(70,'deg')
```

H =

```
0.3420    -0.9397    4.0000
0.9397     0.3420    9.0000
0          0         1.0000
```

```
>> H1 = eye(3)
```

```
H1 =
```

```
    1    0    0
    0    1    0
    0    0    0
```

```
>> tranimate2(H,H1)
```

TRANIMATE(P1, P2, OPTIONS) animates a 3D coordinate frame moving from

pose X1 to pose X2. Poses X1 and X2 can be represented by:

- SE(3) homogeneous transformation matrices (4×4)
- SO(3) orthonormal rotation matrices (3×3)

TRANIMATE(X, OPTIONS) animates a coordinate frame moving from the identity pose to the pose X represented by any of the types listed above.

TRANIMATE(XSEQ, OPTIONS) animates a trajectory, where XSEQ is any of

- SE(3) homogeneous transformation matrix sequence ($4 \times 4 \times N$)
- SO(3) orthonormal rotation matrix sequence ($3 \times 3 \times N$)

```
>> H = transl(1,2,3) * troty(45,'deg')
```

```
H =
```

```
    0.7071    0    0.7071    1.0000
         0    1.0000         0    2.0000
   -0.7071         0    0.7071    3.0000
         0         0         0    1.0000
```

```
>> H1 = eye(4)
```

```
H1 =
```

```
    1    0    0    0
    0    1    0    0
    0    0    1    0
    0    0    0    1
```

```
>> tranimate(H,H1)
```

Try out

1. Prove the orthogonal property of a rotation matrix. Hint: use determinant and inverse function in MATLAB.
2. In a three angle rotation representation singularity occurs. Prove it with an example.